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Pratheeksha Sumesh

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EDUCATION

Mahindra University (École Centrale School of Engineering) <i>B.Tech. Computational Mathematics</i>	Expected: May 2028 <i>Current GPA: 8.33/10.0</i>
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COURSEWORK

Courses: Optimization, Data Structures & Algorithms, Linear Programming, C Programming, Statistics, Calculus, Linear Algebra, Probability Theory
Awards: Selected as Best Project at Computational and Mathematics Symposium (2025)

SKILLS

Languages: C, Python, SQL (learning), R
Tools: Git/GitHub (learning), Jupyter Notebook, VS Code, Excel, NumPy, Pandas, scikit-learn
Mathematical: Linear Programming, Optimization Modeling, Statistical Analysis, Algorithm Design

PROJECTS

ML Data Preprocessing Pipeline — Breast Cancer Dataset <i>Python, Pandas, NumPy, scikit-learn</i> <i>Machine Learning Pipeline Developer (Group Project, MA2221 · Mahindra University)</i>	Jan. 2026 – May 2026
- Built end-to-end data preprocessing pipeline in Python on the 569-sample, 30-feature Wisconsin Breast Cancer Dataset, extracting feature matrix X and target vector y, validating zero missing values, and computing summary statistics using Pandas and NumPy. - Implemented and compared manual 80/20 train-test split via NumPy index shuffling against sklearn's <code>train_test_split</code>, verifying class balance consistency (train: 62.6%, test: 63.2%) as individual contribution to Section 4 of group project. - Applied StandardScaler fitted exclusively on training data to standardize 30 continuous features to zero mean and unit variance, correctly preventing data leakage ahead of downstream classification.	
India Multidimensional Poverty Index Analysis <i>R, Python, Statistical Analysis, ML</i> <i>Statistical Analyst — Model Comparison Lead (Group Project, MA2220 · Mahindra University)</i>	Jan. 2026 – May 2026
- Benchmarked 5 statistical and ML models predicting multidimensional poverty across 707 Indian districts (NFHS-5 data): Linear Regression ($R^2=0.49$), Logistic Regression, Decision Tree ($R^2=0.83$, 84.5% accuracy), and Random Forest ($R^2=0.96$, 92.3% accuracy) using R and Python. - Identified literacy rate and years of schooling (10+) as dominant poverty predictors in Random Forest (importance: 0.148 and 0.569 respectively), and child marriage and youth-population share as strongest linear regression drivers, directly informing team's policy recommendations. - Evaluated accuracy vs. interpretability trade-off across model classes and authored the model comparison section, recommending ensemble models for national-level targeting and linear models for policy communication.	
Meal Planning Optimization <i>Python, NumPy, Linear Programming, Optimization</i> <i>Optimization Developer & Researcher</i>	Apr. 2025
Formulated meal planning optimization problem for institutional dining facilities using linear programming and dual simplex method to minimize costs while meeting nutritional requirements for 500+ students. Implemented dual simplex algorithm in Python leveraging NumPy and optimization libraries through collaborative 4-person team effort, delivering functional meal planner interface. Selected by professor as best project from course for presentation at Computational and Mathematics Symposium in August 2025, competing against 20+ teams.	

- **Developed** stock market analysis program in C to evaluate **20+ companies** based on **7 financial metrics** including P/E ratio, revenue, net profit, market cap, debt-to-equity, dividend yield, and promoter holding.
- **Implemented** multi-parameter scoring system using **structs, arrays, and sorting algorithms** to rank companies and generate **investment recommendations** with percentage allocations based on comparative analysis.
- **Designed** efficient data processing logic handling **140+ data points** to provide **actionable investment insights** through automated ranking system.

Python & Machine Learning Certification | Python, Scikit-learn, Pandas, NumPy, ML
Machine Learning Student

Dec. 2024 – Mar. 2026

- **Pursuing** 3-month Python with Machine Learning certification program at Unlox, gaining **hands-on experience** in **supervised/unsupervised learning, neural networks**, and data preprocessing using Python libraries.
- **Applying** machine learning algorithms including **regression, classification, and clustering** to analyze real-world datasets and address **business case scenarios** across retail, finance, and healthcare domains.

EXPERIENCE

Entrepreneurship Project | Team Leader & Financial Manager

Jan. 2025 – May 2025

Business Operations Manager

- * **Led 5-member team** in launching jewelry and embroidery business with **Rs. 5,000 seed capital**, managing end-to-end operations including **pricing strategy, vendor negotiations, product procurement**, and financial accounting to achieve **100% ROI (Rs. 10,000 revenue)** within 5 months.
- * **Developed and executed** multi-channel go-to-market strategy across Instagram, college campus pop-ups, and word-of-mouth marketing, generating **50+ customer orders** and establishing **repeat customer base of 15+ clients**.
- * **Maintained** detailed financial records using **Excel**, tracking revenue, expenses, and profit margins to ensure profitability and inform pricing decisions.

Model United Nations - UNHRC | Delegate of Russia

2023

Diplomatic Negotiation Leader

- * **Represented Russia** as delegate in UNHRC committee at interschool Model United Nations conference with **100+ participants**, lobbying and negotiating with **15+ country delegates** to pass **3 resolutions** on global human rights issues.
- * **Led coalition of 8 supporting nations** through strategic lobbying, persuasion, and diplomatic negotiations, successfully merging two smaller blocs into unified alliance representing **40% of committee voting power** to advance policy objectives.

INTERESTS

Professional Interests

Data-driven business analytics, optimization modeling, machine learning applications, financial market analysis, public speaking and presentations

Personal Interests

International travel and cultural immersion, music, exploring emerging AI technologies